# Lecture 6

2/2/26

## Resolution

(Ch. 4 in text)

There were two big pioneers in the field of diffraction-limited resolution

Lord Rayleigh - astronomer, studied self-luminous objects
Abbe - microscopist, discuss later

Microscopes have lots of apertures, including lens openings.
Apertures diffract light, causing it to spread.

Ray optics - just trace rays, which land at image point, no spreading

Wave optics - light spreads, degrading images

Objects that are too close together, or viewed w/ a system having apertures that are too small, can't be distinguished due to diffraction

Will see that wavelength also affects amount of diffraction, & thus resolution, w/ longer wavelengths (red) worse than shorter wavelengths (blue)

Why not just use blue light in μscopy?

- Sensors more sensitive to red (& even more sensitive to infrared)
- Blue fluorophores bleach faster
- Multicolor imaging needs several colors (typically red, green, blue)
- Don't normally use UV because it kills cells

<u>Key implications</u> of diffraction

(1) Resolution improves as $\begin{cases} D\uparrow & \text{Aperture diameter increases} \\ \lambda\downarrow & \text{Wavelength decreases} \end{cases}$

(2) Point objects are imaged as a diffraction pattern

Small objects are distorted in size & appear too large

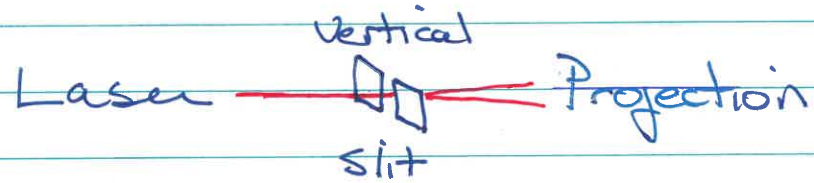
Limit on resolution ≈ 200 nm = 0.2 μm
(w/ <u>blue</u> light)

Objects that are smaller than about 0.2 μm in <u>radius</u> appear enlarged to 0.2 μm in radius

<u>Example</u>: synaptic vesicles have a radius of 0.05 μm but appear to be at least 0.2 μm in light μscope

(3) Objects more closely spaced than ~0.2 μm cannot be distinguished as two objects but instead blur into one object

**DEMO** Diffraction

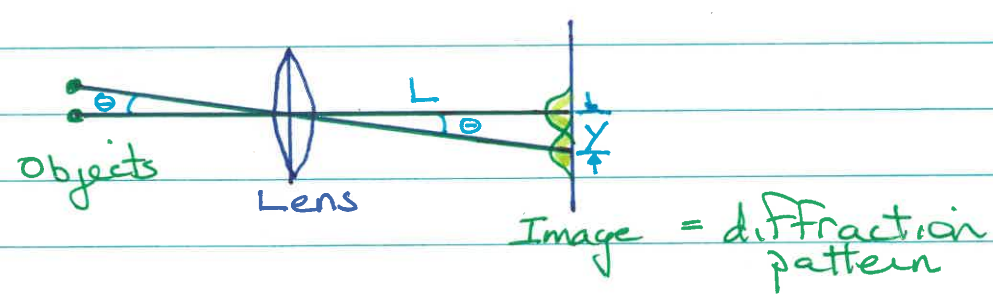



Results:

- No slit → spot
- Wide slit → spot
- Narrow slit → elongated diffraction pattern

**Rayleigh Theory** Hard optics, focus on results

Quantifies angular resolution




How close in angle can two objects get & still be resolved?

wavelength in vacuum (or air)

$$\tan\Theta_{min} = \frac{y}{L} = \boxed{\frac{1.22\lambda_0}{D} \approx \Theta_{min}}$$

small angle approximation

Minimum resolvable angular separation (in radians)

This is what happens w/ eyes

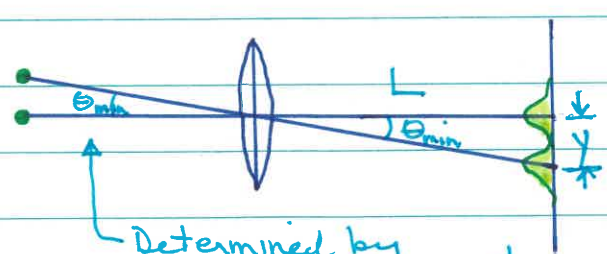
Want $\Theta_{min}$ to be as small as possible

Key takeaways from equation $\begin{cases} \lambda\downarrow & \Theta_{min}\downarrow \\ D\uparrow & \Theta_{min}\downarrow \end{cases}$

Two other things are important

(a) What is the minimum resolvable spatial separation $y_{min}$ between two objects?

(b) What is the maximum distance $L_{max}$ from sensor (e.g., eye) two objects can be & still be resolvable?



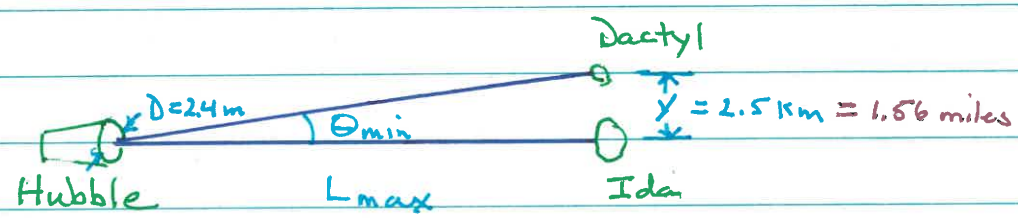

$$\boxed{\tan\Theta = \frac{y}{L}}$$

Determined by system parameters ($\lambda_0$ & D)

(a) Solve for $y_{min}$ if L is fixed & known

(b) Solve for $L_{max}$ if y is fixed & known

# ACTIVITY 2 Diffraction-Limited Resolution




- Calculate $\theta_{min}$ from $\lambda$ & D

$$\theta_{min} = \frac{1.22\lambda}{D} = \frac{1.22 \cdot 550 \cdot 10^{-9}\,m}{2.4\,m} = 2.8 \cdot 10^{-7}\,rad$$

- Calculate $L_{max}$ from $\theta_{min}$ & y

$$\theta_{min} = \frac{y}{L_{max}} \Rightarrow L_{max} = \frac{y}{\theta_{min}} = \frac{2.5 \cdot 10^{3}\,m}{2.8 \cdot 10^{-7}\,rad} = \boxed{8.9 \cdot 10^{9}\,m}$$

$= 5.56 \cdot 10^{6}$ miles

The Hubble aperture is large $\begin{cases} \text{Increased resolution} \\ \text{Increased light detection} \end{cases}$

~~Hubble distinguishes two objects about 1½ miles apart from about 6½ million miles away~~

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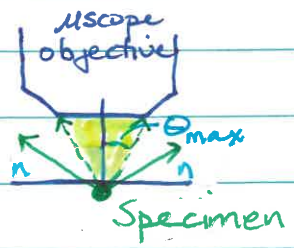
Focus to this point on "aperture"

Abbe, who worked for Zeiss, first to focus on "<u>numerical</u> aperture" (NA)




$$\boxed{NA = n \sin\theta_{max}}$$

n = refractive index of medium between specimen & objective

$\theta_{max}$ = half-angle of cone of collected light

The medium between the specimen & objective is typically one of the following:

| | |
|---|---|
| Air | $n = 1$ |
| Water | $n = 1.33$ |
| Oil | $n = 1.52$ |

(Increasing ↓)

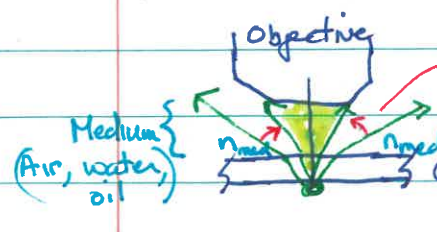
A large NA is good in μscopy { More light collected; Better resolution (next lecture) }

NA increases as n increases & $\theta_{max}$ increases

$n \uparrow \; NA \uparrow$

$\theta_{max} \uparrow \; NA \uparrow$

SIMULATION  Mol Expressions  Immersion Oil & Refractive Index




Light bends away from objective & is NOT captured by objective if $n_{med} < n_{glass}$

$n_{oil}$ generally matches $n_{glass}$ & so maximizes light collection

For a μscope, the minimum resolvable separation, $d_{min}$, is given by

$$\boxed{d_{min} = \frac{0.61\,\lambda_o}{NA}}$$

Want $d_{min}$ <u>small</u>

$$d_{min} \downarrow \begin{cases} NA \uparrow & \text{Want large NA} \\ \lambda_o \downarrow & \text{Want short wavelength} \end{cases}$$

This equation can be used to derive a useful rule of thumb about μscope resolution

$$\text{Assume} \begin{cases} NA \approx 1 & \text{(NAs can get up to } \sim 1.5) \\ 0.61 \approx 0.5 & \\ \lambda = 400\text{ nm} & \text{(use lowest } \lambda \text{ visible light)} \end{cases}$$

$$d_{min} \approx \frac{0.5 \cdot 400\text{nm}}{1} \approx 200\text{ nm}$$

Abbe theory will confirm results

Start next lecture

6-8

Name _____

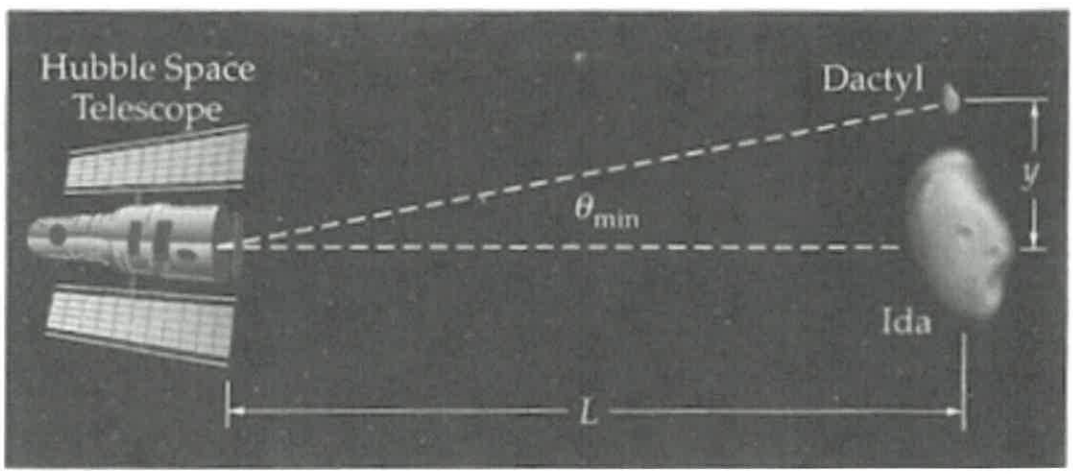
# Physics 390 In-Class Activity 2

## *Diffraction-limited Resolution – Rayleigh's Angular Version*

As discussed in class, Lord Rayleigh suggested that Airy diffraction patterns from two objects be regarded as barely resolvable if the angular separation of the objects is at least as big as $\theta_{\text{min}}$, given by

$$\tan\theta_{\text{min}} = \frac{y}{L} = \frac{1.22\lambda}{D} \backsim \theta_{\text{min}}$$

Here $D$ is the diameter of the diffracting aperture, $y$ is the linear separation of the objects, and $L$ is the distance of the objects from the aperture (see image below).




The asteroid Ida is orbited by the moon Dactyl. If the separation between Ida and Dactyl is 2.5 km, what is the maximum distance at which they can be resolved by the Hubble Space telescope (diameter = 2.4 m) using 550 nm light?